# BUS-F420: Equity and Fixed Income Investments

## Lecture 13: Measuring Bond Yields

**Prof. Derek Wenning**

March 4, 2026

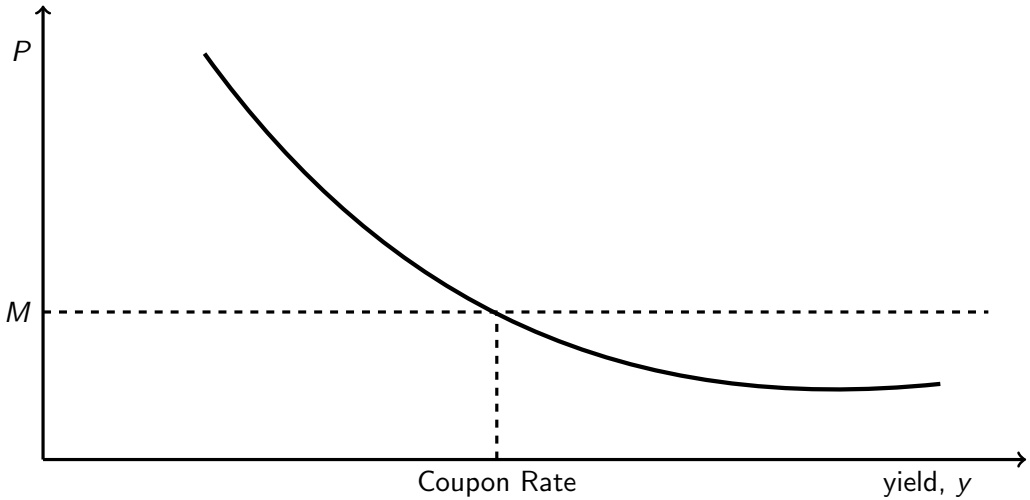
# Recap from last time

- Last time, talked about how to value a bond

- **Key mechanisms:**
  1. Lower maturities $\rightarrow$ higher prices *if selling at a discount*
  2. Embedded (call) options $\rightarrow$ lower prices to compensate
  3. Higher market yields $\rightarrow$ lower prices to compensate

- **Today:** Given the **price** of a bond, what is its implied **yield**?
  $\rightarrow$ Closer to an idea of its *return* given its *cost*

# Recap on Price-Yield Relationship



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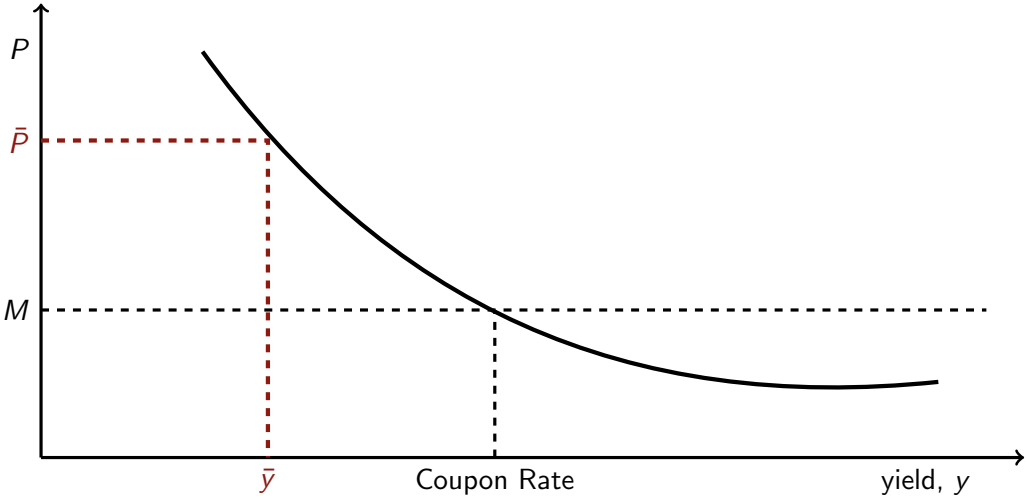

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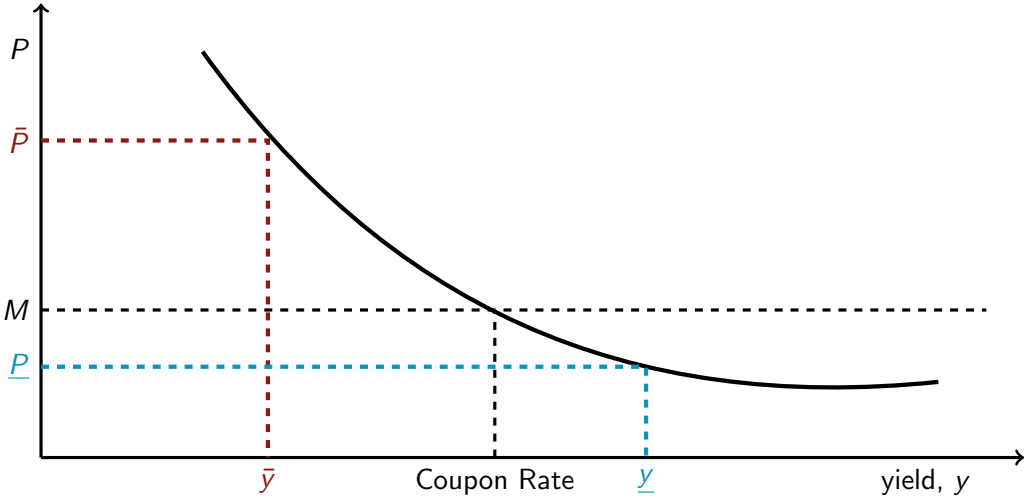

# Why/when do we care about prices vs. yields?

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- Although the two are intimately connected, they serve different purposes

- **Prices** reflect value, which matters for *asset holdings*
  - Ex 1: Banks must report *risk-based capital*, which is derived from asset valuations
  - Ex 2: Retirement accounts

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- **Yields** reflect income relative to costs, which matters for *returns*
  - Allows us to make comparisons across relatively *dissimilar* bonds
  - **Today:** what information is contained in a yield?

# What do prices tell us about yields?

- **Question:**
  *If we hold a bond to maturity*, what is the *yield* we should receive that is consistent with a given price?

- Call this the **yield to maturity**, which solves

$$\text{(Given!)} \rightarrow \boldsymbol{P} = \sum_{t=1}^{T} \frac{CF_t}{(1+\boldsymbol{y})^t} \leftarrow \text{(Needs to be determined!)}$$

# Calculating Yield to Maturity and Bond Prices in Excel

- **Challenge:** We can't *directly* solve for **yield to maturity** $\rightarrow$ use Excel!

- **Key Function:** `RATE(periods, coupon, −price, par value)`
  - `periods`: # of payment periods remaining in the bond
  - `coupon`: size of the coupon payment ($r \times M$)
  - `price`: current price of the bond — **must be entered negatively**
  - `par value`: par value of the bond

# Calculating Yield to Maturity and Bond Prices in Excel

- Let's work through an example using a 5-year corporate bond where:
  - (Annual!) Coupon rate ($r$): 8%
  - Price: \$900
  - Par value: \$1,000

- How should we fill out our Excel function?

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- How should we fill out our Excel function?

**Formula:** $\texttt{RATE}(\texttt{periods}, \texttt{coupon}, \texttt{-price}, \texttt{par value})$

**Our bond:** $\texttt{RATE}(10, 40, -900, 1000) = \mathbf{5.315\%}$

## Let's practice with the following bonds:

`RATE(periods, coupon, -price, par value)`

| Years Remaining | Annual Coupon Rate | Market Price | Par Value | **YTM** |
|---|---|---|---|---|
| 8 | 10% | \$850 | \$1,000 | |
| **6** | 10% | \$850 | \$1,000 | |
| 8 | **8%** | \$850 | \$1,000 | |
| 8 | 10% | **\$800** | \$1,000 | |

(Remember to add decimal places to your computed rates!)

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| 8 | 10% | **\$800** | \$1,000 | **7.14%** |

(Remember to add decimal places to your computed rates!)

# What information is imbedded in yield to maturity?

- **So far:** Two key characteristics determine yields — coupon rates + capital gains
- But let's rearrange our equation for yield to maturity:

$$\underbrace{(1+y)^T - 1}_{\substack{\text{required return} \\ \text{(total yield to maturity)}}} =$$

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- **Imbeds a key assumption: reinvestment of coupon payments!**

# How important is each channel?

$$(1+y)^T - 1 = \frac{TC}{P} + \frac{M-P}{P} + \frac{C\sum_{t=1}^{T} R_{T-t}(y)}{P}$$

| Years Left | Annualized Coupon Rate | Market Price | Par Value | *Total* YTM | Current Yield | Capital Gains | Reinv. Yield |
|---|---|---|---|---|---|---|---|
| 8 | 10% | \$850 | \$1,000 | 175.5% | | | |
| 6 | 10% | \$850 | \$1,000 | 122% | | | |
| 8 | 8% | \$850 | \$1,000 | 133% | | | |
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|---|---|---|---|---|---|---|---|
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| 6 | 10% | \$850 | \$1,000 | 122% | 70.6% | 17.7% | 33.9% |
| 8 | 8% | \$850 | \$1,000 | 133% | 75.3% | 17.7% | 40.0% |
| 8 | 10% | \$800 | \$1,000 | 201% | 100% | 25% | 76.3% |

## What are the limits to yield to maturity?

- Clearly, reinvestment is an important part of yields!
- **Q: What are some issues with how we've treated reinvestment income?**

# What are the limits to yield to maturity?

- Clearly, reinvestment is an important part of yields!
- **Q: What are some issues with how we've treated reinvestment income?**
  - Assumed a *consistent* reinvestment rate!
- **True yields will likely reflect reinvestment risk**
  - Depend on *how* and *when* you reinvest!

# Portfolio Yields

- So far, we've talked about a **single** bond. But what about a **portfolio?**

  Suppose you hold the following bonds (market value: $2,617):

| Bond | Years | Annual Coupon Rate | Market Price | Par Value | YTM |
|---|---|---|---|---|---|
| A | 1 | 8% | $945.75 | $1,000 | 7% |
| B | 2 | 7% | $850.75 | $1,000 | 8% |
| C | 3 | 10% | $820.50 | $1,000 | 9% |

- Q: What do you think the yield to maturity of your portfolio should be?

# Portfolio Yields

| | Cash Flow | | | | (Discount rate: 8.000%) |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |

True portfolio value: 2617

# Portfolio Yields

(Discount rate: 8.000%)

| | Cash Flow | | | | |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | 40 | | | | |
| 2 | 1040 | | | | |
| 3 | 0 | | | | |
| 4 | 0 | | | | |
| 5 | 0 | | | | |
| 6 | 0 | | | | |

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# Portfolio Yields

(Discount rate: 8.000%)

| | Cash Flow | | | | |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | 40 | 35 | | | |
| 2 | 1040 | 35 | | | |
| 3 | 0 | 35 | | | |
| 4 | 0 | 1035 | | | |
| 5 | 0 | 0 | | | |
| 6 | 0 | 0 | | | |

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# Portfolio Yields

(Discount rate: 8.000%)

| | Cash Flow | | | | |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | 40 | 35 | 50 | | |
| 2 | 1040 | 35 | 50 | | |
| 3 | 0 | 35 | 50 | | |
| 4 | 0 | 1035 | 50 | | |
| 5 | 0 | 0 | 50 | | |
| 6 | 0 | 0 | 1050 | | |

True portfolio value: 2617

# Portfolio Yields

(Discount rate: 8.000%)

| Period | Cash Flow | | | Total Cash Flow | Discounted Cash Flow |
|---|---|---|---|---|---|
| | Bond A | Bond B | Bond C | | |
| 1 | 40 | 35 | 50 | 125 | |
| 2 | 1040 | 35 | 50 | 1125 | |
| 3 | 0 | 35 | 50 | 85 | |
| 4 | 0 | 1035 | 50 | 1085 | |
| 5 | 0 | 0 | 50 | 50 | |
| 6 | 0 | 0 | 1050 | 1050 | |

True portfolio value: 2617

# Portfolio Yields

| | Cash Flow | | | | (Discount rate: 8.000%) |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | 40 | 35 | 50 | 125 | 115.74 |
| 2 | 1040 | 35 | 50 | 1125 | 964.51 |
| 3 | 0 | 35 | 50 | 85 | 67.48 |
| 4 | 0 | 1035 | 50 | 1085 | 797.51 |
| 5 | 0 | 0 | 50 | 50 | 34.03 |
| 6 | 0 | 0 | 1050 | 1050 | 661.68 |

True portfolio value: 2617

# Portfolio Yields

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|---|---|---|---|---|---|
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True portfolio value: 2617

**Value: 2640.94**

# Portfolio Yields

| | Cash Flow | | | | (Discount rate: 8.272%) |
|---|---|---|---|---|---|
| Period | Bond A | Bond B | Bond C | Total Cash Flow | Discounted Cash Flow |
| 1 | 40 | 35 | 50 | 125 | 115.45 |
| 2 | 1040 | 35 | 50 | 1125 | 959.67 |
| 3 | 0 | 35 | 50 | 85 | 66.97 |
| 4 | 0 | 1035 | 50 | 1085 | 789.53 |
| 5 | 0 | 0 | 50 | 50 | 33.60 |
| 6 | 0 | 0 | 1050 | 1050 | 651.78 |

True portfolio value: 2617 **Value: 2617**

# Yield to Maturity Useful for Fixed Rate But...

- As with computing **prices**, several characteristics come to our aid:
  1. Fixed cash flows
  2. Fixed discount rates
  3. Fixed maturity

- How can we use our **yield to maturity** framework to evaluate
  - bonds with embedded options?
  - bonds with floating rates?

# Yield to Maturity of a Callable Bond

- Let's consider a bond that's **callable** at some point before maturity
  - Not callable *yet*
  - Symbolically, call this time $T_C$ $(< T)$

- **Q: What's the challenge we face, even given $T_C$?**

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  - We don't know if the issuer will actually call!
  - Depends on market prices, which are uncertain

- **So what should we do?**

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  - We don't know if the issuer will actually call!
  - Depends on market prices, which are uncertain

- **So what should we do? A: Worst case scenario — yield to call**

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(Example: 5 year corporate with par $1000 that's callable in one year at par)

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- **Same as YTM!** Just replace:
  (1) "`periods`" $\rightarrow$ "`periods until callable`"
  (2) "`par value`" $\rightarrow$ "`strike price`"

- **Q: Based on our previous exercises, how should this affect yields?**

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  - A: They should increase *given* the price of a bond!

- But wait... last time we said callability $\rightarrow$ lower prices. **How do we rationalize this?**
  - Prices $\downarrow$ if callable *given yields!*

# Call Schedules

- So far been a bit imprecise about callable bonds
  - Assumption 1: Call price = par value
  - Assumption 2: Single call date in the future

  **Reality:** Callable bonds often have a *call schedule*

- Q: Suppose a schedule has multiple call dates. How might you expect call prices to vary over time?

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  **Reality:** Callable bonds often have a *call schedule*

- Q: Suppose a schedule has multiple call dates. How might you expect call prices to vary over time?
  - **A: They typically decline!**
  - Want to *discourage* early calls

# Example 1: Call Schedules and Yield to Call, Discount Bond

- Let's return to a 10-year corporate bond with:
  - Annual coupon rate 8%, price $950, par value $1,000

| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | |
| 4 | $1,025 | |
| 8 | $1,000 | |

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| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | 9.48 |
| 4 | $1,025 | |
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| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | 9.48 |
| 4 | $1,025 | 5.14 |
| 8 | $1,000 | 4.44 |

- **Q: Which one is the "relevant" yield?**
  **A:** The lowest — "yield to worst" — is standard practice in industry

# Example 2: Call Schedules and Yield to Call, Premium Bond

- Let's return to a 10-year corporate bond with:
  - Annual coupon rate 8%, price $1,100, par value $1,000

| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | |
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  - Annual coupon rate 8%, price $1,100, par value $1,000

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|---|---|---|
| 1 | $1,050 | 1.379 |
| 4 | $1,025 | |
| 8 | $1,000 | |

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| 4 | $1,025 | 2.866 |
| 8 | $1,000 | |

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| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | 1.379 |
| 4 | $1,025 | 2.866 |
| 8 | $1,000 | 3.192 |

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- Let's return to a 10-year corporate bond with:
  - Annual coupon rate 8%, price $1,100, par value $1,000

| Years to call | Call Price | Yield to Call |
|---|---|---|
| 1 | $1,050 | 1.379 |
| 4 | $1,025 | 2.866 |
| 8 | $1,000 | 3.192 |

- **Q: Which one is the "relevant" yield?**
  **A:** The 1-year call date!

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$$\text{Return} = \underbrace{\frac{C}{P}}_{\substack{\text{fixed}\\ \text{cash flow}}} + \underbrace{\frac{\Delta P}{P}}_{\substack{\textbf{uncertain}\\ \textbf{cash flow!}}}$$

$\rightarrow$ Must **project** how a bond's value changes to understand <u>*intermittent*</u> returns

- **Next Time:** How do changes in yields affect bond returns?